

[illegible]

1. Triad: Uosm>100 + low urate + euvolemic = SIADH

WHAT YOU SEE

Top banner: 'Uosm over 100, low uric acid, euvolemic equals SIAD'.

WHAT IT ENCODES

SIADH is euvolemic HYPOtonic hyponatremia from inappropriate ADH: serum osm <275, urine osm >100 (usually >300), urine Na >30–40, clinically euvolemic, with low uric acid (<4) and low BUN from dilution + urate washout. Normal thyroid and adrenal function are required — it's a diagnosis of exclusion. Mnemonic: euvolemic + concentrated urine + low urate = SIADH.

BOARD TRAP

Don't call it SIADH before excluding hypothyroidism and adrenal insufficiency — both cause euvolemic hyponatremia that looks identical, and cortisol replacement (not fluid restriction) fixes adrenal cases.

2. SIAD Hospital (setting)

WHAT YOU SEE

Banner: 'SIAD Hospital — Syndrome of Inappropriate Antidiuresis'.

WHAT IT ENCODES

ADH (vasopressin) is released independent of osmolality, so the kidney retains free water, dilutes the serum, and produces concentrated urine. Water is retained but Na is not — total body water rises slightly while the patient stays clinically euvolemic. The result is dilutional hyponatremia, the most common cause of hyponatremia in hospitalized patients.

BOARD TRAP

SIADH is a water problem, not a salt-deficit problem — giving salt alone or normal saline can paradoxically worsen the Na if urine osm is high (desalination).

3. Urine osm >100 (often >300)

WHAT YOU SEE

Yellow sign 'Uosm over 100'.

WHAT IT ENCODES

In SIADH the kidney can't dilute urine despite a hypotonic serum, so urine osm is inappropriately >100 mOsm/kg (commonly >300). This is the single key lab proving ADH is active when it shouldn't be.

BOARD TRAP

If urine osm is <100, ADH is appropriately suppressed — that's primary polydipsia or low-solute (beer potomania / tea-and-toast), NOT SIADH.

4. Urine Na >30–40

WHAT YOU SEE

Yellow sign 'U Na over 30'.

WHAT IT ENCODES

Because the patient is euvolemic (even mildly volume-expanded), the kidney excretes sodium normally, giving urine Na >30–40 mEq/L. This separates SIADH from hypovolemic hyponatremia.

BOARD TRAP

A low urine Na (<20) means the kidney is avidly retaining sodium → think true volume depletion, not SIADH (unless the patient is also salt-restricted).

5. Low uric acid (<4) & low BUN

WHAT YOU SEE

Yellow sign 'Uric acid less 4'.

WHAT IT ENCODES

Mild volume expansion increases urate and urea clearance, so SIADH classically shows hypouricemia (<4 mg/dL) and a low BUN — supportive dilutional markers.

BOARD TRAP

A HIGH uric acid / high BUN points to volume depletion (hypovolemic hyponatremia), not SIADH — don't ignore the urate.

6. Euvolemic — normal volume

WHAT YOU SEE

Banner 'Euvolemic normal volume'.

WHAT IT ENCODES

No edema, no ascites, no orthostasis or dry mucosa — the exam is normal. AVP retains water but ongoing natriuresis prevents frank volume overload, so the patient is clinically euvolemic.

BOARD TRAP

Edema/ascites = HYPERvolemic hyponatremia (HF, cirrhosis, nephrotic); dry/orthostatic = HYPOvolemic — both are managed differently from SIADH, so always assess volume first.

7. Copeptin/AVP NOT for diagnosis

WHAT YOU SEE

Central sign 'No copeptin — X diagnosis; plasma AVP/copeptin not diagnostic'.

WHAT IT ENCODES

SIADH is confirmed by the osmolality/urine/volume pattern, not by measuring AVP. Plasma AVP and copeptin overlap too much with other states to be diagnostic and aren't part of the routine criteria.

BOARD TRAP

Ordering an AVP or copeptin level to 'prove' SIADH is the wrong move — the diagnosis is made from serum osm, urine osm, urine Na, volume status, and normal thyroid/adrenal/renal function.

8. SCLC — #1 malignancy cause

WHAT YOU SEE

Bottom-left bed labeled 'SCLC — 75% of malignancy'.

WHAT IT ENCODES

Small cell lung cancer is the classic ectopic ADH producer and the most common malignant cause of SIADH (paraneoplastic). New SIADH in a smoker should prompt a chest CT.

BOARD TRAP

SQUAMOUS cell lung cancer causes hypercalcemia via PTHrP — not SIADH; don't swap the two paraneoplastic syndromes.

9. Pulmonary — PNA, TB, abscess

WHAT YOU SEE

Bed labeled 'Pulmonary — PNA, TB, abscess'.

WHAT IT ENCODES

Lung infections (pneumonia, TB, lung abscess) and positive-pressure ventilation are common pulmonary triggers of SIADH. Legionella pneumonia in particular is associated with hyponatremia.

BOARD TRAP

Hyponatremia with a 'pneumonia' that also has diarrhea and high fever + relative bradycardia points to Legionella — don't treat with a beta-lactam alone (needs a macrolide/fluoroquinolone).

10. CNS — meningitis, SAH, trauma

WHAT YOU SEE

Bed labeled 'CNS — meningitis, SAH, trauma'.

WHAT IT ENCODES

CNS disease (meningitis, subarachnoid hemorrhage, head trauma, stroke, tumors) is a major SIADH cause. SIADH is treated with fluid restriction.

BOARD TRAP

In SAH/CNS injury, distinguish SIADH (euvolemic/hypervolemic → fluid-restrict) from cerebral salt wasting (HYPOvolemic → give salt + saline) — the treatments are opposite, and over-restricting a CSW patient worsens cerebral ischemia.

11. Post-op, pain, nausea

WHAT YOU SEE

Bed labeled 'Postop / nausea'.

WHAT IT ENCODES

Pain, nausea, and the post-operative state are powerful NON-osmotic stimuli for ADH release. Combined with hypotonic IV fluids, they drive hospital-acquired hyponatremia.

BOARD TRAP

Running D5W or half-normal saline in a post-op/nauseated patient is a classic cause of iatrogenic hyponatremia — use isotonic (0.9% saline/LR) for maintenance instead.

12. SSRIs

WHAT YOU SEE

Right bed with an 'SSRI' sign.

WHAT IT ENCODES

SSRIs (and SNRIs) are among the most common drug causes of SIADH, especially in the elderly within the first weeks of starting. Check a sodium a few weeks after initiation in older patients.

BOARD TRAP

Attributing an elderly patient's confusion/hyponatremia to 'old age' instead of a recently started SSRI misses a reversible cause — stop the drug.

13. Carbamazepine — thirst + AVP

WHAT YOU SEE

Sign 'Carbamazepine — causes thirst plus AVP'.

WHAT IT ENCODES

Carbamazepine and oxcarbazepine potentiate ADH action and stimulate thirst, a frequent drug cause of SIADH. Other culprits: antipsychotics, MDMA/ecstasy, and desmopressin.

BOARD TRAP

MDMA (ecstasy) hyponatremia is a double hit — SIADH plus large free-water intake; treat the acute severe symptomatic case with hypertonic saline, not just fluid restriction.

14. Chemo — cyclophosphamide

WHAT YOU SEE

Bottom-right bed 'Chemo induced'.

WHAT IT ENCODES

Cyclophosphamide (especially high-dose IV), vincristine, and cisplatin cause SIADH. With IV cyclophosphamide the standard pre-hydration can precipitate dangerous hyponatremia.

BOARD TRAP

Pushing extra free water to 'protect the bladder' during IV cyclophosphamide can worsen SIADH hyponatremia — use isotonic fluids and watch the sodium.

15. Step 1: fluid restriction <1 L/day

WHAT YOU SEE

Treatment ladder rung 'Fluid restrict / salt restrict'.

WHAT IT ENCODES

First-line for chronic, mildly symptomatic SIADH is fluid restriction to <800–1000 mL/day. Add oral salt tabs ± a loop diuretic to raise free-water excretion if restriction alone is insufficient.

BOARD TRAP

Fluid restriction FAILS when urine osm is very high (>500) or when urine (Na+K) exceeds serum Na (negative electrolyte-free water clearance) — recognize this and escalate rather than restricting harder.

16. Urea (osmotic free-water loss)

WHAT YOU SEE

Ladder rung 'Urea promotes free water excretion'.

WHAT IT ENCODES

Oral urea 15–30 g/day induces an osmotic diuresis that excretes free water and raises sodium; it's effective, cheap, and low-risk for overcorrection (palatability is the main limit).

BOARD TRAP

Don't dismiss urea as obsolete — guidelines favor it (and salt + loop) over demeclocycline, which is nephrotoxic and slow.

17. Tolvaptan / vaptans (V2 antagonist)

WHAT YOU SEE

Ladder rung 'Tolvaptan — V2 antagonist'.

WHAT IT ENCODES

Vaptans block the V2 receptor → pure aquaresis (free water out, Na retained), raising serum Na. Start in hospital, do NOT fluid-restrict during initiation, limit tolvaptan to <30 days (hepatotoxicity), and it's expensive.

BOARD TRAP

Combining a vaptan with fluid restriction, or using it for chronic mild SIADH, risks rapid OVERcorrection → osmotic demyelination; reserve it and monitor Na closely.

18. Demeclocycline — NOT recommended

WHAT YOU SEE

Top sign 'Demeclocycline' with a red X / 'not recommended'.

WHAT IT ENCODES

Demeclocycline induces nephrogenic DI (blocks ADH at the collecting duct) but is now discouraged: slow onset, nephrotoxic, and unpredictable. Largely replaced by urea and vaptans.

BOARD TRAP

Picking demeclocycline as a go-to SIADH drug is the wrong answer on modern exams — choose fluid restriction → salt/urea → vaptan instead.

19. Severe/symptomatic → 3% saline

WHAT YOU SEE

Top-right 'Acute-rescue station — 3% saline 100 mL over 10 min, severe neuro symptoms, target 4–6 mEq/L'.

WHAT IT ENCODES

For severe SYMPTOMATIC hyponatremia (seizures, obtundation, coma), give 3% hypertonic saline 100–150 mL boluses to raise Na ~4–6 mEq/L over the first few hours, then stop. Total correction must stay <8 mEq/L in 24 h.

BOARD TRAP

Correcting chronic hyponatremia too fast (>8 mEq/L/24h) causes osmotic demyelination syndrome (central pontine myelinolysis) — if you overshoot, re-lower with D5W ± desmopressin.

20. SGLT2 inhibitor (emerging)

WHAT YOU SEE

Top sign (marked NEW) 'SGLT-2 inhibitor induces glycosuria → osmotic diuresis raises Na'.

WHAT IT ENCODES

Empagliflozin causes glucosuria and an osmotic diuresis that excretes free water, raising serum Na — an emerging adjunct for SIADH in trials. Watch for euglycemic DKA and genital infections.

BOARD TRAP

It's adjunctive/emerging, not first-line — fluid restriction (± salt/urea) remains the initial chronic-SIADH treatment.